Practical value of sequence-based click prediction

The goal is to develop a novel method for click-through rate prediction from user behavior sequences. We will identify a concrete limitation in existing models, design and implement a method that addresses it, and evaluate it against strong baselines. The intended contribution is better prediction quality or a better accuracy-cost trade-off.

Who would use it

An advertising platform or online retailer must rank many eligible items during a short request. Its ranking team needs accurate click estimates while keeping memory, latency and maintenance costs within its serving budget. The project aims to deliver an implemented method that a team could integrate into its CTR model. If validated, the method could improve how the model uses past behavior. Users may receive more relevant recommendations, while advertisers may spend less on poorly matched impressions.

Data and validation

We will have access to Avito internal logs for method development and evaluation. Public datasets support reproducible comparisons. Avazu, Criteo and raw Taobao Ad Click record ad displays and whether each display led to a click. Amazon Electronics records product reviews; Taobao UserBehavior records shopping actions. For training, researchers can use a reviewed or clicked item as a positive example. They can randomly choose another item as a negative example. The datasets do not tell us whether the user saw the randomly selected item. A negative label therefore does not establish that the user saw the item and did not click. The percentage of positive examples depends on how many random negative examples the researchers create. The EDA also checks user and item identifiers, event timing and usable history length.

Feasibility and the main risks

The project is feasible with access to Avito internal logs and sufficient compute resources. The main risks are that improvements in offline metrics may not translate into a positive online A/B test, or that technical constraints may prevent us from running the test. Several reviewed CTR papers use online A/B tests to demonstrate practical value. We therefore consider a positive online result important evidence for the paper we aim to write. Data availability is a low risk given access to Avito logs. Older public datasets pose a moderate risk because user behavior may have changed. A leak of user activity would be a serious security risk. Ethical risk is moderate because more clicks do not necessarily mean more useful recommendations.

How a client would measure quality

Evaluate standard prediction metrics, such as log loss and AUC, offline on logged data. Estimate business metrics offline under conditions that approximate production. Then measure the impact on business metrics in an online A/B test.

Research value

The intended research result is an original sequence-based CTR method, a reproducible implementation and controlled evidence about its benefits and limits. The paper should explain which limitation the method addresses, why its mechanism helps, where it fails and what it costs to train and serve.